Yu (Jennie) Zheng

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SUMMARY

Target: Research Engineer or Privacy/Security Engineer.
Research: AI privacy and large language model (LLM) agent security; private graph learning and secure ML, with publications at NDSS, NeurIPS, and ACSAC.
Privacy & Security: LLM security: build frameworks for agent action authorization and LLM-aided intrusion detection. AI privacy: develop privacy-preserving ML frameworks with differential privacy (DP), DP-SGD, privacy accounting, membership-inference auditing, secure multiparty computation (MPC) and secret sharing.
ML and Evaluation: Built Python benchmarking pipelines and PyTorch/PyG graph neural network (GNN) training/inference; released private-ML code and evaluated synthetic data and federated learning.
Skills: Python, C++, PyTorch; AWS Cloud, Git, Linux, Docker; Codex, Claude Code.

EDUCATION

Chinese University of Hong Kong | Ph.D., Information Engineering *2018 - 2024*

Northeastern University, China | B.Eng., Communication Engineering *2014 - 2018*

EXPERIENCE

University of California, Berkeley *Feb 2026 - Present*

Associate Specialist and Privacy Engineer, Data Privacy Lab, CEGA Berkeley, CA

Outcomes: Global Privacy Playbook | KITA paper | OWL benchmark

- **LLM Agent Security · KITA:** Validated rejection of insufficient quorum, mismatched digests, action substitution and replay in 6 scripted system tests; compared 4 authorization designs across 7 controlled scenarios. Developed a Python LLM-agent authorization prototype binding reviewer approvals to canonical actions through authenticated gateways and threshold BLS verification before execution; kept signing keys outside LLM inputs and verified key-free reviewer inputs (NeurIPS 2026 InfPriv Workshop).
- **Privacy Evaluation · OWL:** Identified classifier- and metric-dependent ranking reversals between differentially private synthetic data and direct private learning, informing task-specific strategy selection. Built a modular Python benchmark comparing AIM, MST and PrivBayes under aligned privacy budgets across 4 public datasets, a 5,100-configuration grid and 864 controlled evaluation cells.
- **Global Privacy Partnerships:** Work with government, NGO and industry partners across countries worldwide to translate privacy needs into protocol frameworks for hierarchical demographic and geographic data; contribute to an emerging Global Playbook and papers.

University of California, Irvine *Sep 2024 - Feb 2026*

Postdoctoral Scholar, Electrical Engineering and Computer Science Irvine, CA

Outcomes: CARIBOU code: github.com/yuzhengcuhk/caribou-public | SHIELD paper | ENTENTE paper

- **Private ML · CARIBOU:** Achieved 85.1% accuracy versus 77.1% for GAP (+8.0 percentage points) on Cora under edge-DP (epsilon = 1). Developed and released a PyTorch/PyG framework for private GNN training/inference with contractive message passing, convergent privacy accounting and membership-inference auditing; evaluated 9 graph datasets with runtime and memory benchmarks (first author, NDSS 2026).
- **LLM-Aided Detection · SHIELD:** Co-authored an LLM-aided host intrusion detection framework achieving 95.8% entity-level precision versus 41.3% for ORTHRUS on DARPA-E3 THEIA. Combined masked-autoencoder log representations, attack-evidence selection/expansion and LLM investigation prompts; evaluated 3 log datasets against 5 HIDS baselines (arXiv, 2025).
- **Federated Detection · ENTENTE:** Co-authored a federated intrusion detector for network log graphs, tested on LANL, OpTC and Pivoting (NDSS 2026); mentored a Ph.D. student on secure graph learning.

Ant Group, Morse Team *May 2022 - Sep 2022*

Applied Research Intern, Algorithm Hangzhou, China

Outcomes: Morse-STF code | ACSAC artifact | Disclosed research covered by patent filings and integrated into products.

- **Secure ML Performance:** Reduced softmax/sigmoid communication by approximately 83%/95% versus prior protocols in published benchmarks; developed secure protocols and contributed CPU/GPU prototypes and MNIST/CIFAR-10 evaluation (co-first author, ACSAC 2023).
- **MPC Implementation:** Authored secure KMeans/PCA components in open-source Morse-STF (Antchain-MPC); used Python shared tensors and batched processing for two-party ML on split data.

Continued research collaboration after internship | VIRGOS, 2025

- **Secure Graph Learning · VIRGOS:** Reduced communication by 50-80% vs. CoGNN and OblivGNN on standard graph datasets; developed two-party GCN training/inference on vertically split graphs via secure sparse matrix decomposition (co-first author; IEEE S&P DLSP Workshop 2025).

2012 Labs, Trustworthy Theory and Engineering Lab *Sep 2020 - Nov 2020*

Applied Research Intern, Security and Privacy Shenzhen, China

SELECTED PUBLICATIONS AND PREPRINTS

992 citations; h-index 11 (Google Scholar, Oct 25, 2026). † Co-first authors.

Conference

[C5] **Y. Zheng**, C. Li, Z. Li, and Q. Wang. "Convergent Privacy Framework for Multi-layer GNNs through Contractive Message Passing." *Network and Distributed System Security Symposium (NDSS), 2026.* [Paper]

[C4] J. Xu, C. Li, **Y. Zheng**, and Z. Li. "Entente: Cross-silo Intrusion Detection on Network Log Graphs with Federated Learning." *Network and Distributed System Security Symposium (NDSS), 2026.* [Paper]

[C3] Z. Yang†, Y. Zhang†, **Y. Zheng**, X. Tian, H. Peng, T. Liu, and B. Han. "FedFed: Feature Distillation against Data Heterogeneity in Federated Learning." *Advances in Neural Information Processing Systems (NeurIPS), 2023.* [Paper]

[C2] **Y. Zheng**†, Q. Zhang†, S. S. M. Chow, Y. Peng, S. Tan, L. Li, and S. Yin. "Secure Softmax/Sigmoid for Machine-Learning Computation." *Annual Computer Security Applications Conference (ACSAC), 2023.* [Paper]

[C1] **Y. Zheng**, W. Song, M. Du, S. S. M. Chow, Q. Lou, Y. Zhao, and X. Wang. "Cryptography-Inspired Federated Learning for Generative Adversarial Networks and Meta Learning." *International Conference on Advanced Data Mining and Applications (ADMA), 2023.* [Paper]

Journal

[J1] X. Ai, Y. Zhu, **Y. Zheng**, G. Li, J. Li, and K. Zhou. "Revisiting Adversarial Robustness of GNNs against Structural Attacks: A Simple and Fast Approach." *IEEE Transactions on Information Forensics and Security (TIFS), 21: 446-459, 2026.* [Paper]

Workshops

[W2] **Y. Zheng**† and Q. Zhang†. "From Review to Authorization: Key-Isolated Threshold Signing for LLM Agents." *Beyond Private Training: The New Landscape of AI Privacy (NeurIPS InfPriv Workshop), 2026.* [Paper]

[W1] **Y. Zheng**†, Q. Zhang†, L. Li, K. Zhou, and S. Yin. "VIRGOS: Secure Graph Convolutional Network on Vertically Split Data from Sparse Matrix Decomposition." *IEEE S&P Deep Learning Security and Privacy (DLSP) Workshop, 2025; arXiv:2502.09808.* [Paper]

arXiv Manuscripts

[M1] D. Sun, J. Zhang, J. Xu, **Y. Zheng**, Y. Tian, and Z. Li. "From Alerts to Intelligence: A Novel LLM-Aided Framework for Host-based Intrusion Detection." *arXiv:2507.10873, 2025.* [Paper]

AWARDS AND PROFESSIONAL SERVICE

ACM MMSys Best Student Paper Award (2022); EPFL SuRI Fellowship (2023).

Selected Program Committee: IEEE S&P, ACM CCS, ACSAC, AAAI, ACNS-SiMLA, ACM WPES@CCS, ICNC.

Reviewer: ACM TOPS, ICLR, ICML, NeurIPS, TMLR, TDSC, AISTATS.